

MATH 3220-002 PSet 5 Specification

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1 Background

This problem set starts with the convergence of infinite series of vectors, and then transitions to differential calculus proper. Recall from MATH 3210 that one can consider infinite series of real numbers as well as infinite series of real valued functions. For both kinds of series, absolute convergence implies convergence, and for function series, **Weierstrass M-test** gives a sufficient condition for the absolute convergence of the sequence of partial sums. First two problems (and perhaps the last problem too) in this problem set generalize these ideas to infinite series of vectors in normed spaces. On the other hand the third and fourth problems focus on derivatives and directional derivatives.

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide

you by providing context and helpful details. The **hyperlinks** too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-

defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook; it's up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. **This problem is a prequel to the next problem.** Let V be a normed space. For $v_\bullet$ a sequence in V , the infinite sum $\sum_{n=0}^{\infty} v_n$ is defined to be the limit of partial sum sequence of $v_\bullet$, if the limit exists, that is,

$$\sum_{n=0}^{\infty} v_n = \lim_{n \rightarrow \infty} \sum_{k=0}^n v_k \in V.$$

In this case we say that the infinite sum $\sum_{n=0}^{\infty} v_n$ **converges**.

$\sum_{n=0}^{\infty} v_n$ is said to be **absolutely convergent** if the infinite sum of scalars $\sum_{n=0}^{\infty} |v_n|$ is convergent.

- (a) V is **Banach** iff for any sequence $v_\bullet$ in V , if $\sum_{n=0}^{\infty} v_n$ is absolutely convergent then it is also convergent.
- (b) If V is Banach, then for any sequence $v_\bullet$ in V , if $\sum_{n=0}^{\infty} v_n$ is absolutely convergent, then

$$\left| \sum_{n=0}^{\infty} v_n \right| \leq \sum_{n=0}^{\infty} |v_n|.$$

2. **This problem is a sequel to the previous problem.** Let E be a Banach space. For any $X \in \mathcal{BL}(E; E)$, consider the formal power series

$$\exp(X) = \sum_{n=0}^{\infty} \frac{X^n}{n!}.$$

- (a) For any $X \in \mathcal{BL}(E; E)$, $\exp(X) \in \mathcal{BL}(E; E)$ is well defined and

$$\|\exp(X)\|_{\text{op}} \leq e^{\|X\|_{\text{op}}}.$$

- (b) For any $X \in \mathcal{BL}(E; E)$, $\exp(X)$ is invertible with

$$\exp(X)^{-1} = \exp(-X).$$

- (c) If $Y \in \mathcal{BL}(E; E)$ is invertible with $Y^{-1} \in \mathcal{BL}(E; E)$, then there is an $X \in \mathcal{BL}(E; E)$ such that $\exp(X) = Y$.

3. **There is nothing to prove in this problem.** Let $M = \{(x, y) \in \mathbb{R}^2 \mid |x| \leq 10, |y| \leq 10\}$ and consider

$$f : M \rightarrow \mathbb{R}^3,$$

$$(x, y) \mapsto (xy - 2x + y + 1, x^2 + y^2 + x - 3y + 6, xy^2).$$

Perform the following tasks:

- (a) Compute the derivative of f . By definition, the derivative of f is a function

$$f' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3).$$

- (b) Compute the derivative of f at $x = (1, -1)$. By definition, the derivative of f at $(1, -1)$ is a linear operator $f'(1, -1) \in \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)$.

- (c) Compute the affine function $A : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that

$$f(1 + h, -1 + k) = A(h, k) + o_{|(h,k)| \rightarrow 0}(|(h, k)|).$$

- (d) Compute the directional derivative of f at $x = (1, -1)$ in the direction of $v = (15, 2)$. By definition, the directional derivative of f at $x = (1, -1)$ in the direction of $v = (15, 2)$ is a vector $D_{(15,2)}f(1, -1) \in \mathbb{R}^3$.

- (e) Compute the derivative of f' . By definition, the derivative of f' is a function

$$(f')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)).$$

- (f) Compute the derivative of $(f')'$. By definition, the derivative of $(f')'$ is a function

$$((f')')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3))).$$

4. Let E be the Banach space of all $n \times n$ matrices with real entries and denote by M the collection of invertible $n \times n$ matrices. Consider the inversion operation defined by:

$$\Phi : M \rightarrow M, \quad X \mapsto X^{-1}.$$

Then

- (a) M is an open subset of E .
- (b) For any $X \in M$ and any $H \in E$,

$$\Phi'(X)H = -X^{-1}HX^{-1}.$$

- (c) Φ is a C^1 diffeomorphism.

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a **generative AI tool** or a **computer algebra system** for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or

- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., [ChatGPT](#)) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the [Wayback Machine](#) (see their "[Save Page Now](#)" documentation) to create a permanent, timestamped archive of your chat. There are also third-party browser extensions that are designed to extract chat logs from generative tools.
- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as [prompt engineering](#), and it is your responsibility you

design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

Here are some guides regarding prompt engineering:

- <https://platform.openai.com/docs/guides/prompt-engineering>
- <https://developers.google.com/machine-learning/resources/prompt-eng>
- <https://www.ibm.com/topics/prompt-engineering>
- <https://aws.amazon.com/what-is/prompt-engineering/>

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/RVxDXkrm7ez9eGtE8>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

<https://www.gradescope.com/courses/1073533/assignments/6451773>,

see the Gradescope [documentation](#) for instructions.

You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

~~This problem set is due on October 3, 2025 at 11:59 PM.~~

This problem set is due on October 6, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.